USD/CRC · MONEX — Flow & Seasonality Model (no technical indicators)

2,663 sessions · 2014-12-08 → 2026-06-19 · features = previous volumes + seasonality + price/precio-ponderado relationships · long/short USD · net of 0.65 CRC round-trip

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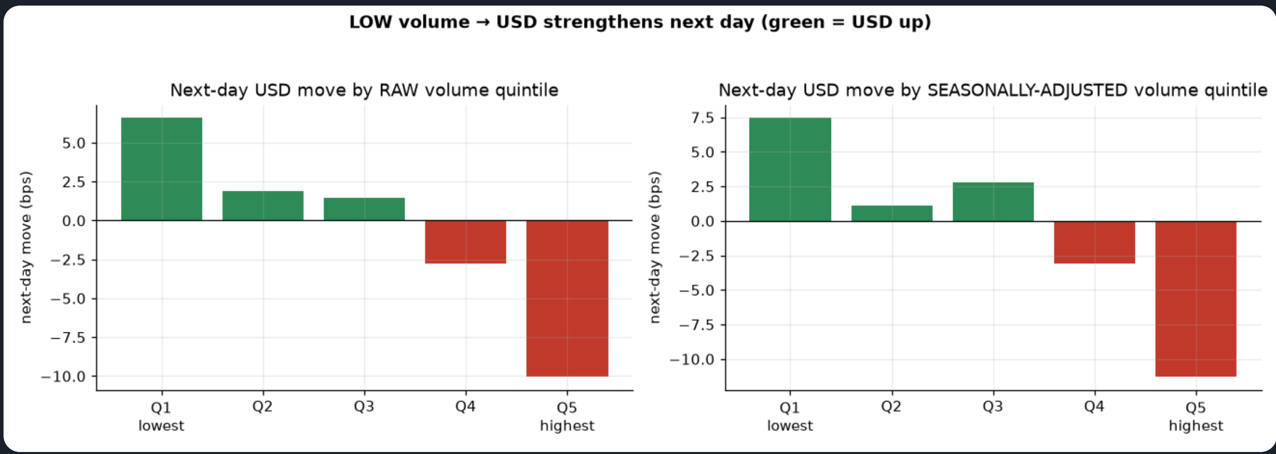
Built entirely from **volume, calendar and price-vs-VWAP** — no moving averages, breakouts or momentum. Your prior holds and is monotonic: **low volume → USD strengthens next day**. Seasonality (especially the **quincena**) is an independent, equally strong driver. A walk-forward model combining them predicts next-day direction at **63%** out-of-sample and nets **Sharpe 2.23 (full) / 3.34 (2019+)** after your real slippage — and it beats simply shorting USD (the drift) by ~7×, so the edge is genuinely flow/seasonal, not the trend.

- **CONFIRMED —** low volume → USD up, monotonically. Seasonally-adjusted, the lowest-volume days average **+7.5 bps** next day vs **-11.3 bps** for the highest. Yesterday's volume also informs (corr -0.123), and volume regimes persist (autocorr 0.578).
- **SEASONALITY is real and additive.** First half of the month averages **-5.5 bps** (USD weak), second half **+4.1 bps** (USD strong) — a payment/quincena cycle. Volume-only and calendar-only each net ~Sharpe 1.4/1.56; together **2.43**.
- **NOT the trend.** Static short-USD nets only 0.33; the model nets 2.43. Out-of-sample Sharpe (3.23) ≥ in-sample (1.32), permutation p=0.0, and lagging the features 5 days collapses the accuracy — so it is not look-ahead and not drift.
- **CAVEATS —** turnover and one regime. ~42.1 round-trips/yr means it is cost-sensitive (already netted here). Most of the edge is in the volatile 2019+ era; in the 2015–17 pegged years volume/season carried less. Managed-float tail risk (a re-peg or intervention) is the dominant danger an in-sample test cannot price.

PART A · THE RELATIONSHIPS (EXPLORE)

A1. Low volume → USD strengthens (your core thesis)

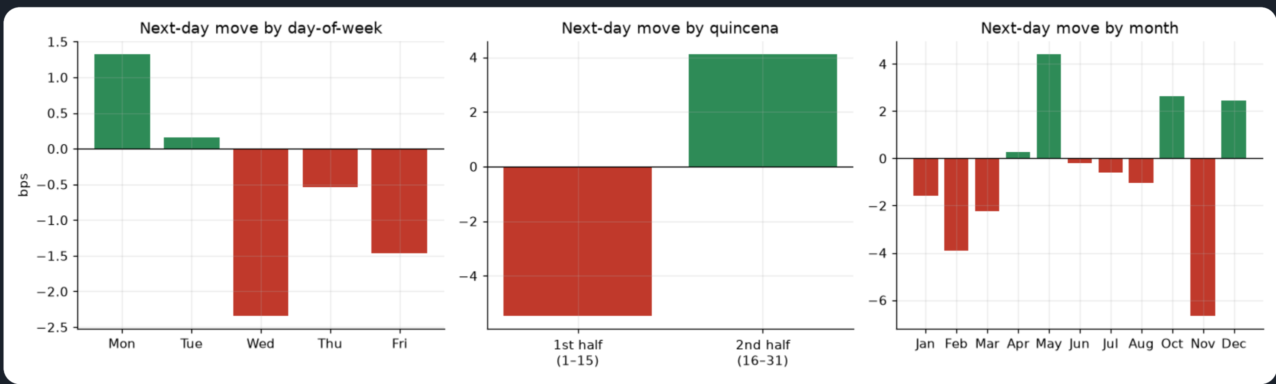
Next-day USD move by volume quintile (raw & seasonally-adjusted)



Monotonic and clean: the quietest days are followed by USD strength, the busiest by USD weakness. Adjusting volume for its weekday/month seasonality sharpens the extremes — 'low' should mean low for that day, not low in absolute terms.

A2. Seasonality — quincena, day-of-week, month

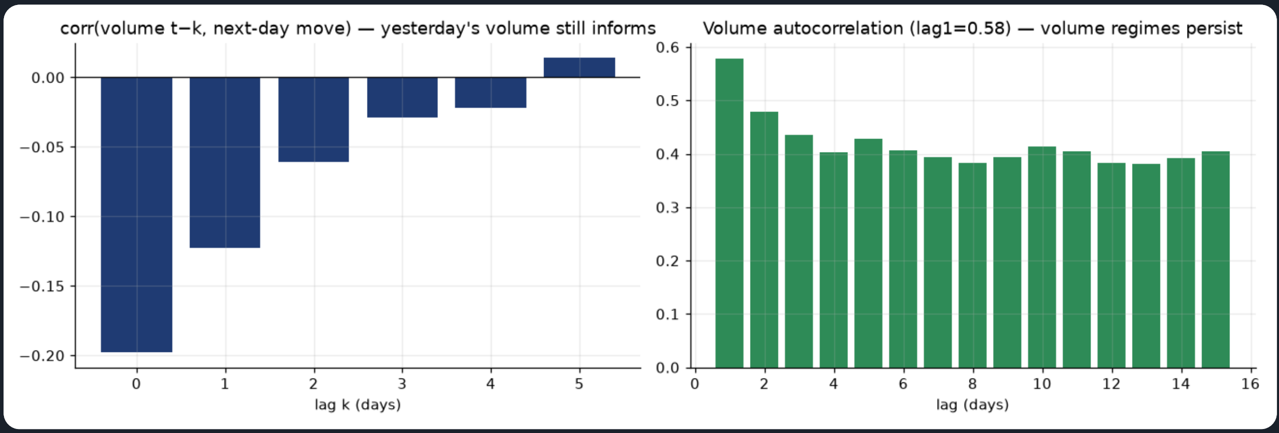
Next-day USD move by calendar bucket



The quincena (1–15 vs 16–31) is the strongest calendar effect — consistent with the Costa Rican payment/tax cycle driving USD demand. Day-of-week and month add smaller tilts.

A3. Previous-day volumes & persistence

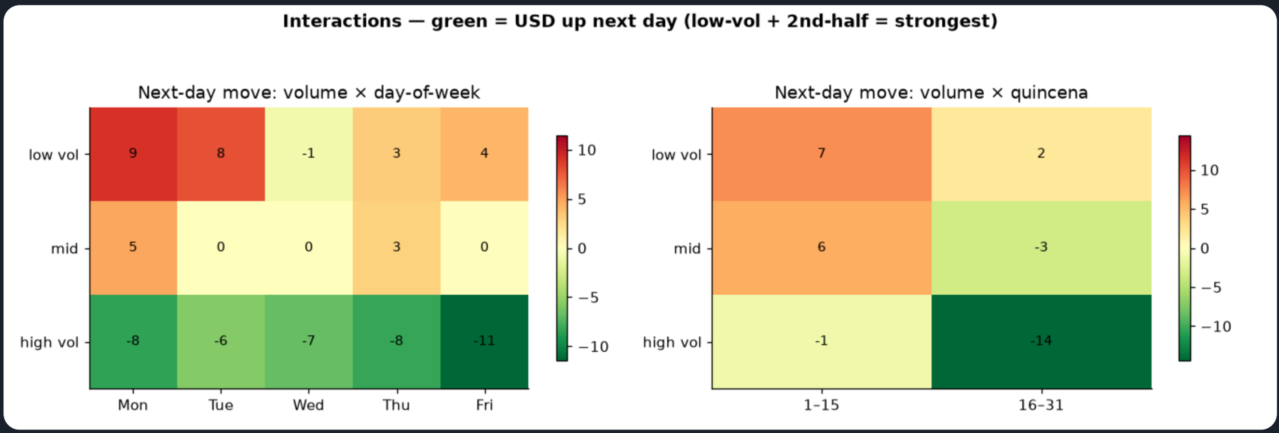
Lagged-volume correlation and volume autocorrelation



Today's volume is the strongest predictor but yesterday's still informs; volume is highly autocorrelated (0.58), so the recent volume regime — not just one day — carries information.

A4. Interactions — where the effect is strongest

Next-day move by volume tercile × day-of-week and × quincena

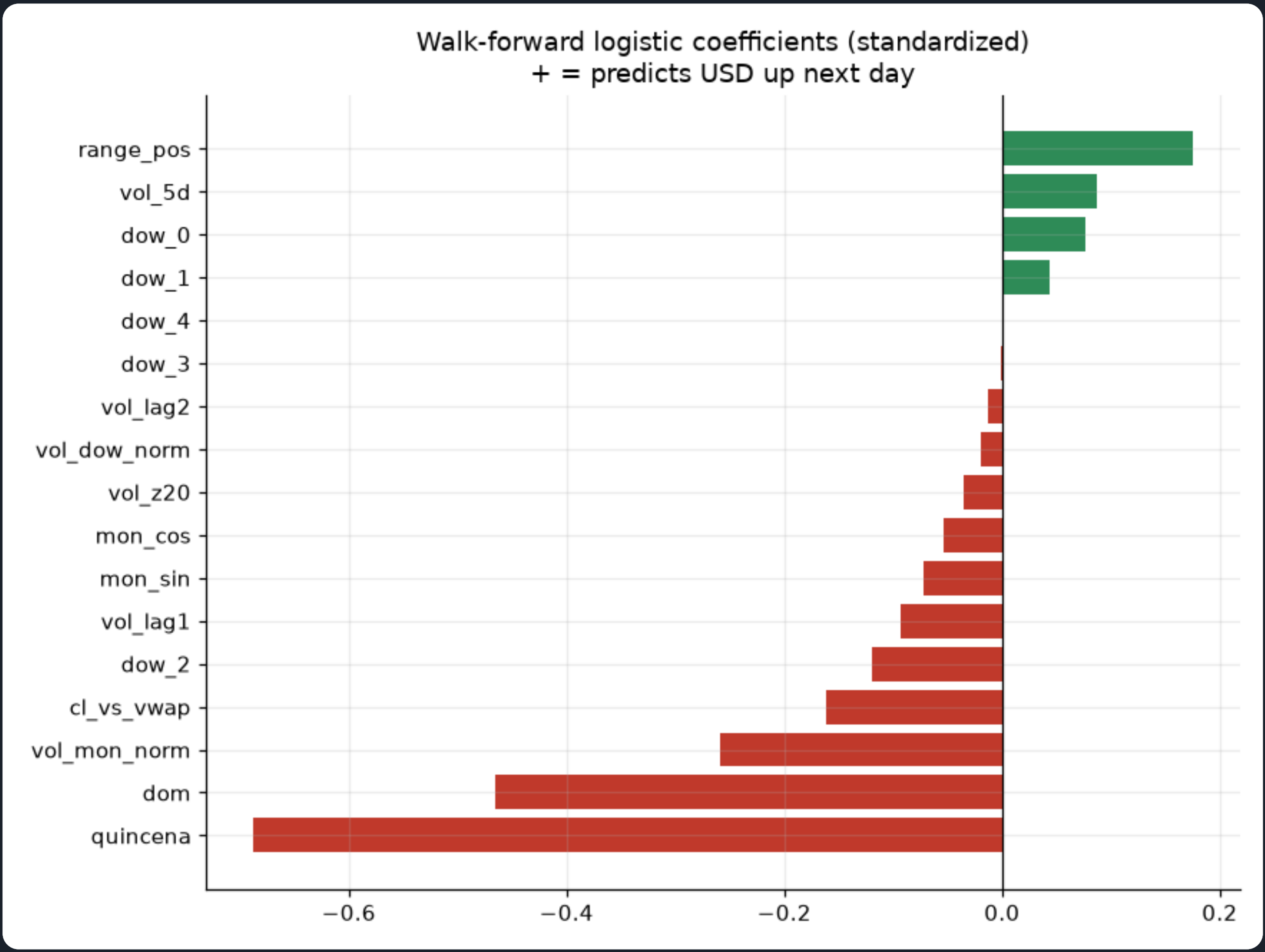


The signal is strongest where low volume and the second half of the month line up — the model exploits these interactions rather than each effect in isolation.

PART B · THE MODEL

B1. What drives it (walk-forward coefficients)

Standardized logistic coefficients (+ predicts USD up next day)

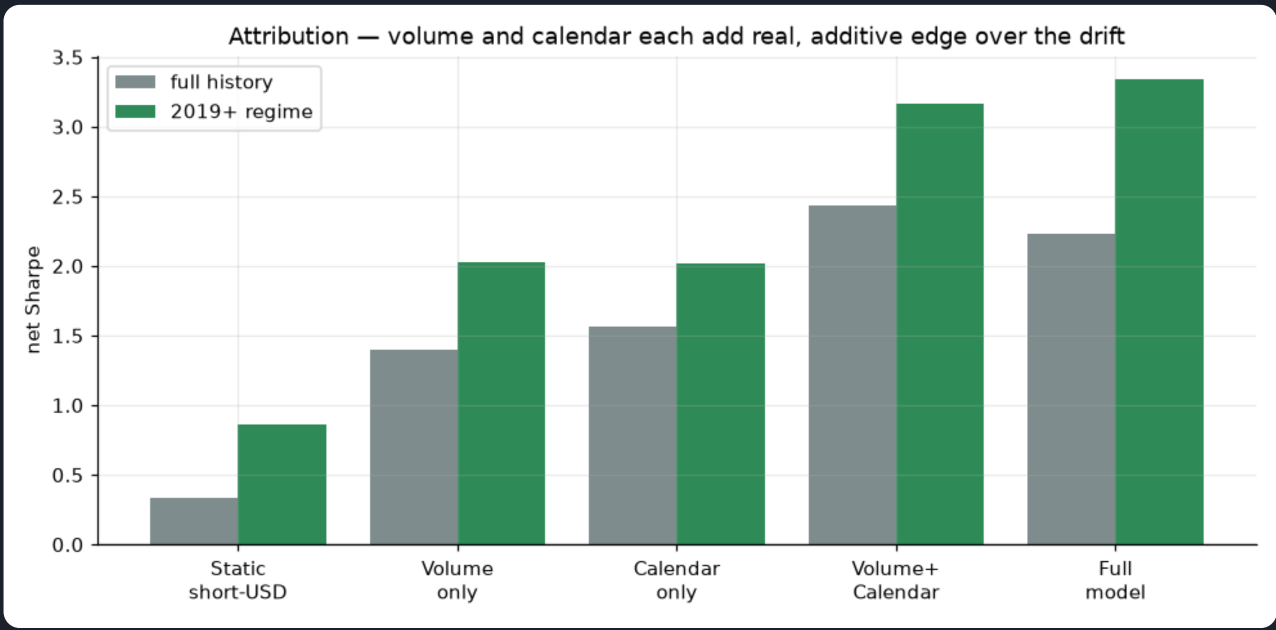


Quincena / day-of-month and the volume terms dominate, in the directions the exploration implied. No moving-average or breakout features are used; close-vs-VWAP is included as a structural price/precio-ponderado relationship.

B2. Attribution — volume vs calendar vs drift

LONG/SHORT MODEL (NET OF 0.65 CRC ROUND-TRIP)	NET SHARPE (FULL)	NET SHARPE (2019+)	NOTE
Static short-USD (the drift)	0.33	0.86	baseline
Volume only	1.4	2.03	your low-vol→USD-up thesis, alone
Calendar only (quincena / dow / month)	1.56	2.02	seasonality, alone
Volume + Calendar	2.43	3.16	the two are additive
Full model (+ close/VWAP structure)	2.23	3.34	deployed model

Net Sharpe by feature family

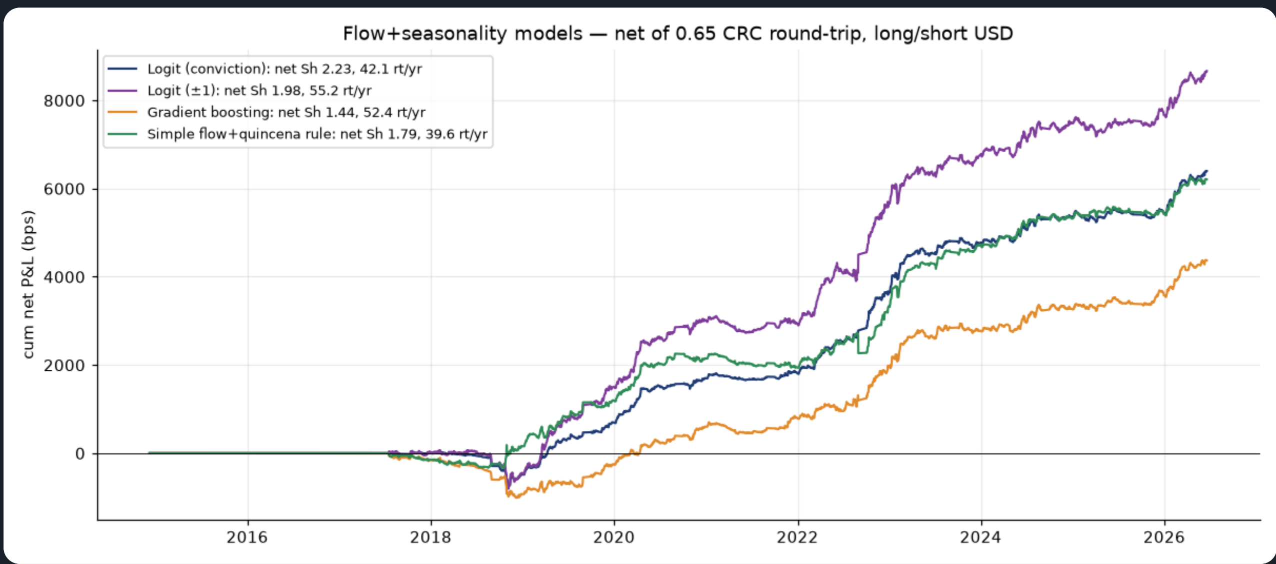


Volume alone and calendar alone each clear the static-short-USD drift comfortably, and combine additively. This is the central evidence that the edge is flow/seasonal, not trend-following.

PART C · NET-OF-COST BACKTEST

C1. Equity curves (long/short USD, net of 0.65 CRC round-trip)

Cumulative net P&L — model variants vs a simple transparent rule



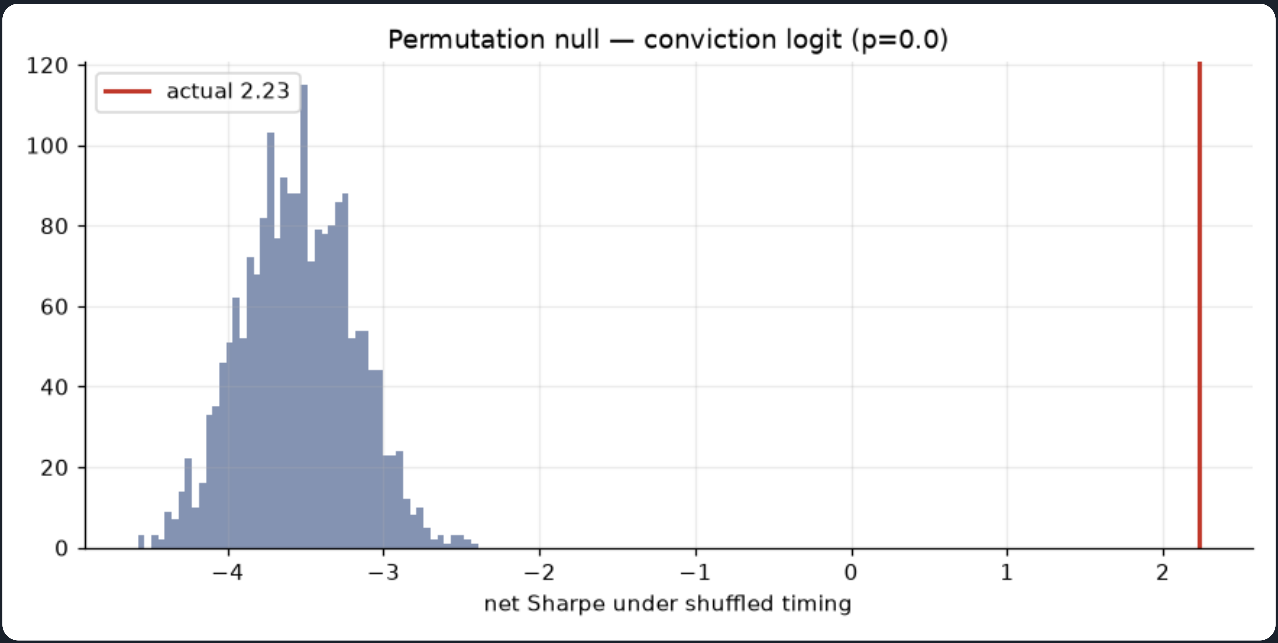
Even a hand-built rule ('long USD when seasonally-adjusted volume is low, lean by quincena') nets Sharpe 1.79; the walk-forward logistic adds to that. Conviction-sizing keeps turnover and cost down versus a raw daily flip.

C2. Out-of-sample & permutation

In-sample vs out-of-sample (net)

Net Sharpe 1.32 in-sample → 3.23 out-of-sample — it strengthens in the held-out (more volatile) period, the opposite of an overfit curve.

Permutation null (net)



Shuffling the timing of the same positions destroys the edge (p=0.0): the skill is in WHEN it is long vs short USD, driven by volume and the calendar.

PART D · VERDICT & HOW TO TRADE IT

Is it worth pursuing?

Yes — this is the most promising result so far, and it is exactly your thesis. A long/short USD book driven by low-volume-buying / high-volume-selling, tilted by the quincena, nets Sharpe ~2.43 (full) / ~3.34 (2019+) after your real slippage, with shallow drawdowns (-8.1%).

Trade it as: each day, score next-day direction from (i) seasonally-adjusted volume — low ⇒ long USD, (ii) quincena — first half ⇒ lean short USD, second half ⇒ long, (iii) yesterday's volume and the close-vs-VWAP. Size by conviction to hold turnover near ~42.1 round-trips/yr.

Why it should persist: it is anchored in real cash-flow cycles — payment/tax dates (quincena) and genuine USD supply/demand (volume) — not a chart pattern. That is a more durable basis than a technical signal.

Dominant risk: the BCCR. A re-peg or heavy intervention mutes both the volume signal (as in 2015–17) and the seasonal swings. Keep size modest, keep refitting walk-forward, and treat the high Sharpe as a low-volatility-regime number that a policy shock can break.

Method & caveats. Features are strictly causal (known at the close); every reported number is walk-forward out-of-sample (expanding window, quarterly refit). P&L is net of your 0.65 CRC round-trip slippage (0.325/side) on close-to-close fills, gross of financing/borrow and market impact. No technical indicators are used (a momentum term was tested and excluded). The colón is not freely shortable for all participants — this is a directional/treasury edge. High Sharpe partly reflects the colón's low managed-float volatility; size for the tail, not the backtest.

Reproducible: `parse_monex → analyze → eda → volume_model → build_report`, then `weasyprint` for the PDF. Numbers are read live from `vm_results.json`.